



Course Name General Physics with Calculus

Course Number PH 212

Course Credits 4

Instructor Dr. Kathryn Hadley

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Class Meeting Times

Lecture: Zoom lectures with optional attendance, schedule to be determined by student poll All in-class assignments and quizzes will have equivalent asynchronous assignments available.
Laboratory: Interactive group meetings schedule to be determined by group availability.

Prerequisite and/or Corequisite

Prerequisite: PH 211, MTH 252

Prerequisite with concurrency: MTH 254

Catalog Course Description

Extends and adapts models based on forces and energy to describe and quantify the fundamental physical phenomena of rotational motion, oscillatory motion, collective motion, and optics. Explores how multiple representations, mathematical sensemaking, and physical explanations strengthen the construction and deployment of the models. Investigates the methods by which physics is built through experiment and observation.

Diversity, Equity and Inclusion

We strive to create an equitable and inclusive community in which all members are welcome, heard, and treated with respect. We uphold these values and take the opportunity to learn from each other. Our greatest strengths and most innovative ideas come from disagreements and collaborations among people with diverse perspectives, lived experiences, and expertise.

Flexibility

We want you to succeed. Flexibility is built into this course as automatically allowed excused absences. Please immediately email the instructor regarding special circumstances or personal issues that arise during the term. We encourage you to contact us if you have concerns about the course, your standing in it, or your ability to complete your work.

Academic Integrity

See the [OSU Code of Student Conduct](#). Collaboration is strongly encouraged on all non-quiz assignments. Your name on a submission signifies that you made a significant contribution to the effort. You should be sure to cite all students you collaborate with on any given assignment and any resources you use.



Communication

Canvas is the source of class information, videos, resources, assignments, and grades. We also send information via announcements with daily announcements made at the start of lecture, so be sure to watch the lecture video if you do not attend the Zoom lecture.

The best way to reach an instructor or TA is via email. We do our best to respond within 24 hours during the week or by Monday over the weekend.

Course Materials and Technology

- Textbook: Physics for Scientists and Engineers, Knight. The 5th edition is used, but any edition will suffice.
- In-class questions and attendance: Poll Everywhere
- Assignment submissions: Gradescope (no subscription required)
- You are required to have a modern browser, web cam and stable internet connection.

Getting Academic Help

- Office hours are online via Zoom after lecture and by appointment.
- The Physics Help Center [Virtual Wormhole](#) is staffed by physics grad and undergrad students.
- TAs assigned to the course area available to help, please see the TA contact information in the Course Information module in Canvas.

Evaluation of Student Performance

- 40%: Quizzes with the lowest quiz score dropped
- 15%: Final exam
- 25%: Lab (You must pass the lab to pass the course)
- 10%: Homework
- 5%: Formative Assessment (Participation)
- 5%: Pre-lecture questions

Example. A student earned 68% on the first quiz (dropped), 75% on the second, 91% on the third, 92% of the homework points, 90% of the formative assessment points, 88% of the pre-lecture points, 92% points for lab, and 82% on the final exam. The grade for the student is:

$$.40(.75 + .91)/2 + .10(.92) + .05(.90) + .05(0.88) + .25(.92) + .15(0.82) = 87\%$$

93 - 100% = A

80 - 82% = B-

68 - 69% = D+

90 - 92% = A-

78 - 79% = C+

63 - 67% = D

88 - 89% = B+

73 - 77% = C

60 - 62% = D-

83 - 87% = B

70 - 72% = C-

0 - 61% = F



Course Content Dates are tentative and subject to change

Week	Date	Chapter	Sections Covered	Exam/Homework	Lab
1	M 6/22	intro			Lab 0: Team Building and Uncertainty Refresher
	T 6/23	4	04.4-6		
	W 6/24	4			
	R 6/25	8	08.1-5	Hw 1 (ch4) due 6/26	
2	M 6/29	8		Proctorio Practice Quiz	Lab 1 Propagation of Uncertainty
	T 6/30	8			
	W 7/1	12	12.1-11		
	R 7/2	12		Hw 2 (ch8) due 7/3	
3	M 7/6	Quiz 1		Quiz 1 (ch 4,8)	Lab 2 Hypothesis Testing
	T 7/7	12			
	W 7/8	12			
	R 7/9	12		Hw 3 (ch12a) due 7/10	
4	M 7/13	12			Lab 3 Data Analysis
	T 7/14	13	13.1-6	Hw 4 (ch12b) due 7/14	
	W 7/15	13			
	R 7/16	15	15.1-8	Hw 5 (ch13) due 7/17	
5	M 7/20	Quiz 2		Quiz 2 (ch 12,13)	Lab 4 Extending a Model
	T 7/21	15			
	W 7/23	16	16.1-9	Hw 6 (ch15) due 7/23	
	R 7/24	16			
6	M 7/27	16			Capstone Lab Design
	T 7/28	17	17.1-8	Hw 7 (ch16) due 7/28	
	W 7/29	17			
	R 7/30	17		Hw 8 (ch17) due 7/31	
7	M 8/3	Quiz 3		Quiz 3 (ch 15,16,17)	Lab 5 Applying a Model
	T 8/4	33	33.1-4		
	W 8/5	33			
	R 8/6	33		Hw 9 (ch33) due 8/9	
8	M 8/10	34	34.1-7		Capstone Lab
	T 8/11	34			
	W 8/12	34		Hw 10 (ch34) Due 8/12	
	R 8/13	Final		Final exam cumulative	

Core Education Category Learning Outcomes

This is a Core Education course that fulfills the requirements for the **Scientific Inquiry and Analysis** category. In addition, this course also fulfills the Baccalaureate Core requirements for the Physical Science category.

Scientific Inquiry and Analysis Learning Outcomes

<i>Outcome</i>	<i>How this outcome will be addressed in the course</i>	<i>How this outcome will be assessed in the course</i>
1. Utilize scientific language, concepts, hypotheses, theories, and laws of basic natural sciences.	<i>Pre-class activities focus on learning fundamental material and practicing key skills using multiple-choice or free-response questions addressing concepts or applications.</i> <i>For example, students will analyze a system using kinematics after having read the appropriate section in the text and watching related videos.</i> <i>Students utilize scientific language and concepts as they are preparing their labs and as they communicate with each other in creating their group report. Students create a hypothesis and address how it relates to the experimental outcomes. They qualitatively compare theory with their actual experimental outcomes and findings in general.</i>	<i>Your solutions to complex quantitative problems, qualitative mechanistic explanations, multiple representations, and reflections are assessed in all of the following assignments:</i> <i>• Pre-lecture assignments</i> <i>• Participation assignments</i> <i>• Weekly homework</i> <i>• Lab group reports</i> <i>• Quizzes and final exam</i> <i>• Group projects</i> <i>Grading criteria address your utilization of scientific language, concepts, hypotheses, theories and laws of basic natural science.</i>
2. Apply the cyclical process of science and think critically by constructing consistent explanations and drawing conclusions based on empirical evidence and current scientific understanding.	<i>In-class activities focus on using concepts, practicing skills, and using physics sensemaking to understand key results. Physics sensemaking techniques that foster thinking critically include unit analysis, covariational reasoning, and special cases.</i> <i>Lab activities focus on setting up experiments, collecting data through observation, analyzing the data by comparing multiple data sets contributed by group</i>	<i>In-class activities are assessed formatively by giving you and your group opportunities to interact with members of the instructional team to receive feedback on the quality of your work.</i> <i>The explanations and conclusions in your lab reports are assessed on your ability: to apply critical thinking about the experiment; to describe the involved physical processes; to</i>



	<i>members for agreement with theoretical expectations (current scientific understanding). A capstone lab project will require students to design their own experiment and write a formal report. This requires them to engage with the cyclical process of science and use critical thinking throughout the project.</i>	<i>report your measurements; to analyze your results, including identification of trends.</i>
3. Articulate the consequences and implications of science for society, daily life, and decision-making	<i>Students are assigned a project where they research a topic related to physics. The project will provide students with an opportunity to connect their learning of physics content to the broader science community and to real-world applications and implications of that content on society. They will gather sources, organize information, investigate implications for daily life, and summarize their findings in the form of a presentation.</i> <i>Students complete intermediate scaffolding assignments in a discussion format, allowing intervention regarding appropriateness and breadth of topic, inclusion of scholarly sources, and organization.</i> <i>Students will make discussion posts to choose a topic and gather and share resources and post their final presentation. Students will then ask substantive questions of other groups' presentations and answer questions posed to them.</i>	<i>Your project will include intermediate assignments and a final presentation. You will be assessed on how well you: describe how your chosen topic relates to social and environmental justice; articulate the consequences and implications of science; and utilize scientific language.</i>

Core Education Category Learning Outcomes

This is a Core Education course that fulfills the requirements for the **Scientific Inquiry and Analysis** category. In addition, this course also fulfills the Baccalaureate Core requirements for the Physical Science category.

Course Specific Measurable Student Learning Outcomes

1. Apply Newton's laws and conservation to analyze the behavior of systems with rotational, oscillatory, or wavelike motion.
 2. Construct a variety of representations, including graphs of related motion quantities, extended free-body diagrams, and symbolic equations, and translate from one representation to another.
 3. Explain physical behavior based on observations, Newton's laws, and conservation principles.
 4. Investigate the physical world to test models and draw conclusions using the scientific method.
 5. Explain how physics principles have led to the development of real-world inventions and devices and how these inventions and devices have impacted society.
- Your learning will be assessed using a wide variety of assignments. Most assignments in this class are **formative**. Your responses to the pre-lecture assignments, participation assignments and homework activities will serve a primary formative role in how you learn physics. Your learning will also be assessed by a term project, labs, quizzes and a cumulative final exam for which you will also receive detailed feedback.

Expectations for Time and Participation

Participation is essential and expected in all components of this course. Attendance in live Zoom lectures is optional, with corresponding asynchronous alternatives available for every synchronous assignment.

This course combines approximately 120 total hours of instruction, online activities, and assignments for 4 credits. This means that, according to OSU policy, students are expected to work on course activities (readings, labs, homework, discussions, assessments, etc.) for approximately 12 hours per week in this 4-credit course. Please plan out your schedule to carve out **at least 12 hours each week** to complete the activities in this course.

This course is asynchronous and somewhat flexible, but not self-paced. Our schedule of Course Content and the due dates that appear in Canvas provide guidelines for how you'll interact and with what frequency. We recommend that you create your own workload schedule and set reminders for assignment due dates.

Textbook: Most students find it helpful to read the textbook before lecture to help understand what occurs during lecture and ask productive questions. Most students also find it helpful to read the textbook after lecture to solidify what they learned during lecture. Think about the questions asked in the book as you go along and make note of what doesn't make sense to you so you can ask about it later. Most importantly, *don't fall behind* because most concepts build on those encountered earlier.

Communication: Communication will be through announcements given in lecture, posted on Canvas, and via email. You are expected to check these daily.

Lecture: Lecture meets online via Zoom for one hour, four times per week. The purpose of lecture will be developing conceptual understanding, working on representing phenomenon, practicing problem solving, and building understanding through observations and explanations of phenomena. Lecture is interactive. There will be times in lecture that you are strongly encouraged to talk and work with other students. Questions, comments, and interruptions are welcome, but please virtually raise your hand. Lecture videos will be posted on Canvas, typically within two hours of the lecture.

Formative Assessment Points: The Poll Everywhere software will be used for formative assessment during lecture. You are allowed three excused absences (three lowest scores dropped) and still earn the maximum points. The three excused absences cover illness, injury, school events, personal days, weather events, work, technical issues, weather, travel, and any other excused reasons. You can alternatively earn participation points via asynchronous assignments delivered in Canvas.

Pre-lecture assignments: Pre-lecture assignments will be open on Canvas for at least 48 hours and will close one hour before class. These assignments will largely be based on assigned textbook reading. Your three lowest scores for pre-lecture assignments will be dropped to cover illness, injury, school events, personal days, weather events, work, technical issues, weather, travel, and any other reasons.

Homework: Online homework will be assigned in Canvas. The lowest homework score will be dropped when calculating the total homework score.

Lab: Lab is the appropriate place for you to apply the tools and skills to explore more complex situations. The labs will get increasingly more open-ended, so eventually



you will be able to do authentic physics modeling of real situations. The lab is fully online with TAs available via Zoom. Each group will submit one lab report. Your percentage of earned points will be included in the calculation of your PH 211 grade. **You must pass lab to pass the course.** You must submit every lab and average at least 70% of the possible points for the lab reports over the course of the term.

Late policy: Lab assignments have a 10% deduction per day late, not to go below 50%. Participation assignments have a deduction of one point per day late. Homework assignments have a deduction of 1 point per day late and are not accepted after solutions are posted. Quizzes, final exam, pre-lecture and extra credit assignments are not accepted late.

Quizzes and final exam: There will be three quizzes and a comprehensive final exam. All quizzes and the final are closed book and comprehensive, including material from readings, labs, lectures, and/or homework. They will consist of conceptual questions and written-out problems and may include problem solving aspects required for homework, such as producing diagrams and/or graphs, discussing assumptions, concepts, and analysis. One quiz score will be excused to accommodate illness, injury, school events, personal days, work, weather events, technical issues, travel, and any other excused reason. You must submit at least three quizzes to pass the course. A quiz reflection assignment will be given for extra credit for the first two quizzes.

Quizzes and the final will be administered in two modalities: proctored via Zoom during the regular class meeting time or at 6pm PDT the same day or an asynchronous alternative. For both modalities, you must have a working web cam. Any student not visible in the web cam during the duration of the quiz will receive a zero for the quiz or exam. The Zoom quiz and final will be administered in Gradescope. The asynchronous quiz and final will be administered via Canvas using Proctorio. The asynchronous questions will be different with the same difficulty level as the Zoom versions, chosen randomly from a question bank in Canvas. For the asynchronous alternative, you need a browser that will support Proctorio and stable internet. If you are dropped from a quiz in Proctorio, it will count as your dropped quiz, and you will be strongly encouraged to attend one of the Zoom meetings for the remaining quizzes. A Proctorio Practice quiz will ensure that you have Proctorio set up correctly before the quiz. You may opt to take a quiz or final exam in an approved testing center. Please contact [Ecampus Testing](#) at least two weeks prior to the event and notify the instructor.



For all quiz and exam modalities, you will be required to sign an integrity statement and submit a scan of an 8.5"x11" notesheet of at least half a page, up to one full page of notes on the current material and your photo ID. The note sheet must be handwritten by you and cannot contain copied images or text or minimized material. Note sheets with AI generated material will not be accepted. No other materials will be allowed during the quiz or exam. Any quiz or exam submitted without an a signed integrity statement and acceptable notesheet with photo ID will receive a score of zero. The note sheet will receive a late deduction of 1 point per day if it is not submitted by the due date/time.

Any official quiz conflict must be discussed, and arrangements made with the instructor *before* the quiz or exam. Unexcused absence will result in a zero for the quiz or exam. If you believe that an error was made in the grading of a question, submit a regrade request in Gradescope *within fifteen days* after graded submissions are posted on Canvas.

Calculators: You will need a calculator for lecture, recitation, lab, and quizzes. If you want to know whether your calculator is acceptable or not for exams, consult the instructor before the first quiz.

Incomplete: To qualify for a grade of Incomplete, you must be missing one major assignment at most (one lab, for example) and be otherwise passing the class with at lease a C-. You need to request an Incomplete from the instructor before the end of the term.

Course Statements

Academic Calendar

All students are subject to the registration and refund deadlines as stated in the Academic Calendar: <https://registrar.oregonstate.edu/osu-academic-calendar>

Statement Regarding Students with Disabilities

Accommodations for students with disabilities are determined and approved by Disability Access Services (DAS). If you, as a student, believe you are eligible for accommodations but have not obtained approval please contact DAS immediately at 541-737-4098 or at <http://ds.oregonstate.edu>. DAS notifies students and faculty members of approved academic accommodations and coordinates implementation of those accommodations. While not required, students and faculty members are encouraged to discuss details of the implementation of individual accommodations."

Student Conduct Expectations link: <https://beav.es/codeofconduct>

Student Bill of Rights

OSU has twelve established student rights. They include due process in all university disciplinary processes, an equal opportunity to learn, and grading in accordance with the course syllabus: <https://asosu.oregonstate.edu/advocacy/rights>

Reach Out for Success

University students encounter setbacks from time to time. If you encounter difficulties and need assistance, it's important to reach out. Consider discussing the situation with an instructor or academic advisor. Learn about resources that assist with wellness and academic success at oregonstate.edu/ReachOut. If you are in immediate crisis, please contact the Crisis Text Line by texting OREGON to 741-741 or call the National Suicide Prevention Lifeline at 1-800-273-TALK (8255)

Student Evaluation of Courses

During Fall, Winter, and Spring term the online Student Learning Experience surveys open to students the Wednesday of week 9 and close the Sunday before Finals Week. Students will receive notification, instructions and the link through their ONID email. They may also log into the system via MyOregonState or directly at beaves.es/Student-Learning-Survey. Survey results are extremely important and are used to help improve courses and the learning experience of future students. Responses are anonymous (unless a student chooses to "sign" their comments, agreeing to relinquish anonymity of written comments) and are not available to instructors until after grades have been posted. The results of scaled questions and signed comments go to both the instructor and their unit head/supervisor. Anonymous (unsigned) comments go to the instructor only.

Academic Integrity: For all quizzes in this course, you will be asked to sign a code of conduct agreement: I understand that in all my OSU courses including this one, I am expected to abide by the Oregon State University "Code of Student Conduct". I recognize that all forms of cheating not only violate the code of conduct I commit to as an OSU student, but they are dishonest, harmful to me and my learning, and unfair to others. By typing my name just below, I certify that on the assignments, quizzes, and exams in this course I will not cheat in any way, including those noted in the "Code of Conduct" excerpt below.

Type your name here in place of a signature: _____

Academic Misconduct. Any action that misrepresents a student or group's work, knowledge, or achievement, provides a potential or actual inequitable advantage, or compromises the integrity of the educational process. Prohibited behaviors include, but are not limited to doing or attempting the following actions:



1. **Cheating.** Unauthorized assistance, or access to or use of unauthorized materials, information, tools, or study aids. Examples include, but are not limited to, unauthorized collaboration or copying on a test or assignment, using prohibited materials and texts, unapproved use of cell phones, internet, or other electronic devices, etc.
2. **Plagiarism.** Representing the words or ideas of another person or presenting someone else's words, data, expressed ideas, or artistry as one's own. Examples include, but are not limited to, presenting someone else's opinions and theories as one's own, using another person's work or words (including unpublished material) without appropriate source documentation or citation, working jointly on a project and then submitting it as one's own, etc.
3. **Falsification.** Fabrication or invention of any information. Examples include, but are not limited to, falsifying research, inventing or falsely altering data, citing fictitious references, falsely recording or reporting attendance, hours, or engagement in activities such as internships, externships, field experiences, clinical activities, etc.
4. **Assisting.** Any action that helps another engage in academic misconduct. Examples include, but are not limited to, providing materials or assistance without approval, altering someone's work, grades or academic records, taking a test/doing an assignment for someone else, compelling acquisition, selling, bribing, paying or accepting payment for academic work or assistance that contributes to academic misconduct, etc.
5. **Tampering.** Interfering with an instructor's evaluation of work by altering materials or documents, tampering with evaluation tools, or other means of interfering.
6. **Multiple submissions of work.** Using or submitting work completed for another or previous class or requirement, without appropriate disclosure, citation, and instructor approval.

Academic Integrity regarding AI

Students are not allowed to post HW assignments to or plagiarize work found on tutoring sites like Chegg and others or to a Generative AI tool. You must submit your own work, thoughts and ideas.

Academic dishonesty will not be tolerated in this class. Students suspected of participating in academic dishonesty **will be reported** to the office of student misconduct and an investigation will be launched.

For more information about academic integrity and the University's policies and procedures in this area, please refer to the [Oregon State University's Code of Student Conduct](#).